

# TAKlite User Guide

## Purpose

TAKlite is a lightweight TAK relay and datapackage service for small ATAK/WinTAK teams over WireGuard VPN.

It provides:

- WireGuard VPN access for admins and TAK clients
- WGDashboard for VPN peer management
- TAKlite admin dashboard for TAK users, clients, and datapackages
- TLS CoT TCP for ATAK/WinTAK on 10.66.66.1:8089
- Plain CoT TCP test port on 10.66.66.1:58087
- Marti-style datapackage upload, search, download, and delete
- Connection User portal for downloading ATAK/WinTAK .dp.zip certificate bundles

## Network Model

TAKlite is designed to be VPN-first.

Public internet:

- 51820/udp WireGuard only
- Temporary 22/tcp SSH only during setup, then close or restrict it

VPN-only:

- 22/tcp SSH
- 10086/tcp WGDashboard
- 8080/tcp TAKlite admin UI, client portal, and HTTP datapackage API
- 8443/tcp TAKlite HTTPS datapackage API
- 58087/tcp plain CoT TCP
- 8089/tcp TLS CoT TCP

## Per-Install Security

Every fresh VPS installation generates new security material.

Do not copy /root/taklite-admin, /etc/wireguard, taklite/certs, taklite/data, or an old .env from one VPS to another.

Generated per install:

- WireGuard server private key
- Initial admin WireGuard peer key and preshared key
- WGDashboard password
- TAKlite bootstrap token
- TAKlite local certificate authority
- TAKlite server certificate
- Generated ATA/WinTAK user certificates

The TAKlite certificate password defaults to atakatak for easier ATA/WinTAK imports unless changed during install.

The installer saves root-only recovery notes here:

```
/root/taklite-admin/README.txt
```

## Fresh VPS Install

Create a fresh Ubuntu 26.04 LTS x64 VPS.

Option A, deploy from a GitHub clone:

```
git clone https://github.com/C137LLC/TAKlite.git
cd taklite
scp -r . root@YOUR_VPS_PUBLIC_IP:/root/taklite
```

On the VPS:

```
cd /root/taklite
chmod +x install.sh smoke-test.sh
./install.sh
```

Option B, deploy from a release zip:

```
scp taklite-vps-bundle.zip root@YOUR_VPS_PUBLIC_IP:/root/
```

On the VPS:

```
cd /root
unzip taklite-vps-bundle.zip
cd /root/taklite-vps-bundle
chmod +x install.sh smoke-test.sh
./install.sh
```

Use the defaults unless the VPS environment requires different values.

Important defaults:

```
WireGuard server: 10.66.66.1
Admin VPN peer: 10.66.66.2
WGDashboard: http://10.66.66.1:10086
TAKlite admin: http://10.66.66.1:8080/
Client portal: http://10.66.66.1:8080/connect/
TLS CoT: 10.66.66.1:8089
Plain CoT: 10.66.66.1:58087
Certificate password: atakatak unless changed during install; saved in `/root/taklite-admin/README.txt`
```

If TAKlite HTTPS/Marti was installed on a custom port, ATAk/WinTAK secure server or datapackage settings should use that custom port.

## Save Install Output

At the end of installation, save:

- Admin WireGuard .conf
- Admin WireGuard QR path
- WGDashboard username/password
- TAKlite bootstrap token
- TAKlite admin URL

Recovery files are saved on the VPS:

```
/root/taklite-admin/README.txt
/root/taklite-admin/bootstrap-token.txt
/root/taklite-admin/admin-wg0.conf
/root/taklite-admin/admin-wg0.png
```

If needed:

```
cat /root/taklite-admin/README.txt
```

## Admin VPN Login

Pull the admin WireGuard config:

```
scp root@YOUR_VPS_PUBLIC_IP:/root/taklite-admin/admin-wg0.conf .
```

Or use rsync:

```
rsync -av root@YOUR_VPS_PUBLIC_IP:/root/taklite-admin/admin-wg0.conf .
```

Import admin-wg0.conf into WireGuard and connect.

Open:

```
WGDashboard: http://10.66.66.1:10086
TAKlite:      http://10.66.66.1:8080/
```

# First TAKlite Admin Account

Open:

```
http://10.66.66.1:8080/
```

Use the bootstrap token from install output or:

```
cat /root/taklite-admin/bootstrap-token.txt
```

Create the first TAKlite admin username/password.

## Create A TAK Device User

Use this workflow for each ATAK/WinTAK device.

- 1 In WGDashboard, create a WireGuard peer for the user device.
- 2 Give the user the WireGuard QR or .conf.
- 3 In TAKlite, open Connection Users.
- 4 Enter a username, temporary password, and optional note.
- 5 Click Create User.
- 6 The user appears in the Connection Users table.
- 7 Use Download DP.zip, QR, or Copy URL.

For bulk onboarding:

- 1 Open Connection Users.
- 2 Click Create Bulk Users.
- 3 Enter a name prefix such as user.
- 4 Enter the number of users to create.
- 5 Click Create Batch.
- 6 TAKlite creates users such as user1, user2, and user3, with matching .dp.zip packages.
- 7 Copy or download the CSV shown after creation. Bulk-created portal users use atakatak as the shared portal password.

Connection User actions:

- Download DP.zip: admin downloads the user's ATAK/WinTAK cert bundle
- Copy URL: copies the VPN-only portal URL
- QR: shows a QR for the user's portal page
- Edit: updates display name and note
- Reset Password: changes the user's portal password

- Allow/Disable Re-download: controls repeated downloads
- Reissue: revokes the old cert bundle and creates a fresh one
- Revoke: disables the user and cert package
- Delete: removes the user record, with optional generated file deletion

## User Device Flow

For the end user:

- 1 Scan/import the WireGuard config.
- 2 Connect WireGuard.
- 3 Open the TAKlite portal QR or URL:

```
http://10.66.66.1:8080/connect/
```

- 1 Log in with the username/password from the admin.
- 2 Download the .dp.zip.
- 3 Import the .dp.zip into ATAK or WinTAK.
- 4 If prompted, use atakatak unless the admin changed the certificate password during install. The active value is saved in:

```
/root/taklite-admin/README.txt
```

The imported package configures:

```
Host: 10.66.66.1
Port: 8089
Protocol: SSL/TLS
```

## ATAK Test

On Android:

- 1 Confirm WireGuard handshake is active.
- 2 Import the .dp.zip in ATAK.
- 3 Confirm the server connection turns green.
- 4 Drop a point.
- 5 Send chat.
- 6 Send a datapackage.
- 7 Confirm other clients receive the point/chat/datapackage.

## WinTAK Test

On Windows:

- 1 Confirm WireGuard is connected.
- 2 Import the `.dp.zip` in WinTAK.
- 3 Confirm the server connection turns green.
- 4 Drop a point.
- 5 Send chat.
- 6 Send a datapackage.
- 7 Confirm ATAK receives it.

## Datapackage Admin

The TAKlite dashboard shows uploaded datapackages.

Admin actions:

- Download datapackage to local machine
- Delete datapackage record and stored file

Expected API activity in logs:

```
POST /Marti/sync/missionupload
GET /Marti/sync/search
GET /Marti/sync/content?hash=...
```

## Connected Clients

The dashboard shows:

- Callsign/name
- UID
- IP address
- Transport mode
- Client certificate common name
- Connection uptime
- Connected time
- Last seen time

TLS clients should show a certificate common name matching the created Connection User or package name.

## Updating TAKlite

Use `update.sh` for normal upgrades. Do not rerun `install.sh` on an existing server unless you are intentionally creating a fresh deployment with new WireGuard and TAKlite credentials.

Updates preserve the live `.env`, so custom ports and network settings stay in place. This includes TAKlite HTTP/HTTPS/CoT ports, WireGuard settings, and the WGDashboard URL.

Release zip workflow:

```
scp TAKlite-vX.Y.Z.zip root@10.66.66.1:/root/

ssh root@10.66.66.1

if [ -f /root/taklite/docker-compose.yml ]; then
    APP_DIR=/root/taklite
elif [ -f /root/TAKlite/docker-compose.yml ]; then
    APP_DIR=/root/TAKlite
elif [ -f /root/taklite-vps-bundle/docker-compose.yml ]; then
    APP_DIR=/root/taklite-vps-bundle
else
    echo "Could not find TAKlite app directory" >&2
    exit 1
fi

bash "$APP_DIR/update.sh" /root/TAKlite-vX.Y.Z.zip
```

Git clone workflow:

```
ssh root@10.66.66.1
cd /root
rm -rf /root/TAKlite-update
git clone https://github.com/C137LLC/TAKlite.git /root/TAKlite-update

bash /root/TAKlite-update/update.sh --from-dir /root/TAKlite-update
```

For a specific release tag:

```
cd /root
rm -rf /root/TAKlite-update
git clone --branch vX.Y.Z --depth 1 \
    https://github.com/C137LLC/TAKlite.git \
    /root/TAKlite-update

bash /root/TAKlite-update/update.sh --from-dir /root/TAKlite-update
```

Never `git clone` directly over the live TAKlite app directory. Clone into a staging directory and let the staged `update.sh` copy application files while preserving `.env`, certs, database, datapackages, and packages.

## Uninstalling Or Reinstalling

Use `uninstall` or `reinstall` only when you intentionally want to wipe the current deployment. Use `update.sh` for normal upgrades.

Warning: both workflows stop WireGuard. If your SSH session is connected through the WireGuard tunnel, the shell can disconnect before the command finishes. Before running either command, make sure public SSH/22 is open from a network you can reach, or have VPS console access ready.

Uninstall wipes TAKlite, WGDashboard, WireGuard config, admin recovery files, certs, users, datapackages, generated packages, and TAKlite backups:

```
cd /root
rm -rf /root/TAKlite-update
git clone --depth 1 https://github.com/C137LLC/TAKlite.git /root/TAKlite-update

bash /root/TAKlite-update/uninstall.sh --yes
```

Reinstall wipes the deployment and starts a fresh install:

```
cd /root
rm -rf /root/TAKlite-update
git clone --depth 1 https://github.com/C137LLC/TAKlite.git /root/TAKlite-update

bash /root/TAKlite-update/reinstall.sh --yes
```

## Troubleshooting

Watch logs:

```
cd /root/taklite-vps-bundle
docker logs -f taklite
```

Check services:

```
docker compose ps
curl -fsS http://10.66.66.1:8080/api/health
wg show
```

Check ports:

```
ss -tulpn | grep -E '(:8080|:8443|:58087|:8089|:10086|:51820)'
```

Common issues:

- Portal login invalid: confirm the user was created under Connection Users, not Connection Packages.
- Invalid truststore: create a fresh Connection User after deploy and import the newest .dp.zip.
- Red TAK connection: confirm VPN is connected, use 10.66.66.1:8089:ssl, and watch logs for cert\_cn=...
- Datapackage send fails: confirm the TAK client has the server connection from the .dp.zip and can reach 10.66.66.1:8080 plus the configured HTTPS/Marti port, usually 8443.

## Smoke Test

From the VPS:

```
cd /root/taklite-vps-bundle
./smoke-test.sh 10.66.66.1 TAKLITE_BOOTSTRAP_TOKEN_OR_ADMIN_TOKEN
```

## Routine Admin Checklist

- 1 Connect admin WireGuard.
- 2 Open WGDashboard.



- 3 Open TAKlite admin dashboard.
- 4 Create WireGuard peer for user.
- 5 Create TAKlite Connection User.
- 6 Give user WireGuard QR.
- 7 Give user portal QR/link and password.
- 8 User downloads/imports .dp.zip.
- 9 Confirm client appears in Connected Clients.
- 10 Test PLI, chat, marker drops, and datapackages.